

Todo list

options (also for packages)	3
indent	3
aumentar longitud	3
piano scores	3
space between notes is weird	3
octaves	4
Ghost note x, hollow note o	5
monotone triplets sliding beam	5
fix clip	5
bar on tuplet, fix height	5
explain better	6
bigger space , ,	6
maybe fix this	7
fix, add swap	9
Fix angle	10
Explain crescendo	10
Half line	10
finish articulation	11
explain overflow	12
explain spaces	12
explain tikz	12

The `mousik` package*

Celia Rubio Madrigal†

July 1, 2020

Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in $\text{\LaTeX}$ because it is based on the `tikz` package.

Keywords

mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key

Contents

1	Introduction	3	3.6.2	<code>\Key</code>	8
			3.6.3	<code>\NoKey</code>	8
			3.7	Other symbols	9
			3.7.1	<code>\Dot</code>	9
2	The <code>mousik</code> environment	3	3.7.2	<code>\Tied</code>	9
2.1	Options	3	3.7.3	The <code>slurred</code> environment . . .	10
3	Available commands	4	3.7.4	The <code>Crescendo</code> and <code>Decrescendo</code> environments . .	10
3.1	<code>\Bar</code>	4	3.7.5	Articulation . . .	10
3.2	<code>\Space</code>	4	3.8	<code>\Text</code> and <code>\Dynamics</code>	11
3.3	<code>\Clef</code>	4	3.9	<code>\Tempo</code>	12
3.4	<code>\TimeSig</code>	5			
3.5	Notes and rests	5	4	Tips and observations	12
3.5.1	<code>\Note</code>	5	5	A very long and complete example	13
3.5.2	<code>\Rest</code>	6			
3.5.3	<code>\Chord</code>	7			
3.6	Keys and accidentals . .	7			
3.6.1	<code>\Acc</code>	7			

*This document corresponds to `mousik` v0.1, dated 2020/07/01.

†Email: celirubio.m@gmail.com

1 Introduction

There are already several ways to set music using L^AT_EX. As stated by the `abc` package, “*notably Music_{TE}X and Lilypond*”, as well as `abc` per se. Some of them are too difficult to use for not too complicated music; others do not work at all.

The `mousik` package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable and lighter packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment.



```
\begin{mousik}
  \Clef
  \Key{-3}
  \TimeSig{4}{4}

  \Rest{8}
  \Note{16}{7,6}
  \Note{|=,-,=|}{5,2,5}
  \Bar

  \Note{1}{5}
  \Chord
  \Acc{0}{9}
  \Note{1}{9}

  \Bar[|. ]
\end{mousik}
```

2.1 Options

options (also for packages)

indent

aumentar
longitud

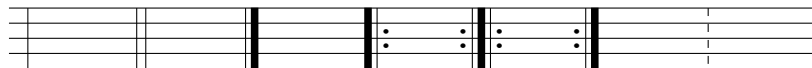
piano scores

space be-
tween notes
is weird

3 Available commands

3.1 `\Bar`

The `\Bar` command prints a bar on the score. It has an optional argument depending on the type of bar, with default `|`, the single bar (`\Bar = \Bar[|]`).

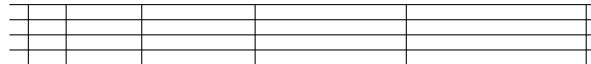


`[|]` `[||]` `[|.]` `[|:]` `[:::]` `[::|]` `[:]` `\Space`

The only way to end a line in the score is with a bar or a space. Any other way may cause visual errors (see subsection 4.1).

3.2 `\Space`

The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number (see subsection 4.2).



`[-1][0]` `[1]` `[2]` `[3]`

3.3 `\Clef`

Clefs might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1). The `\Clef` command has an optional argument which corresponds to its type; the Treble Clef is the default (`\Clef = \Clef[2]`).

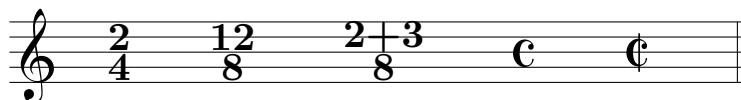
octaves



`[1]` `[2]` `[3]` `[4]` `[11]` `[12]` `[13]` `[14]` `[15]` `[0]`

3.4 `\TimeSig`

Time signatures might be used at any moment. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets. Also `\TimeSig{C}{}` will produce a *common time* symbol and `\TimeSig{C|}{}` will produce an *alla breve* symbol.

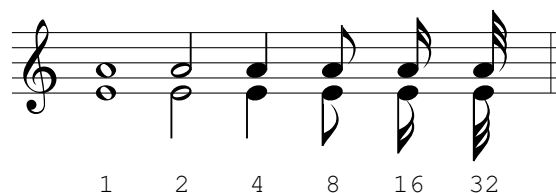


3.5 Notes and rests

3.5.1 `\Note`

The `\Note` command prints a note on the score. It accepts 2 arguments.

The first argument is the duration of the note, from one of these:



Ghost note
x, hollow
note o

The second argument is the absolute height. For example, these are the notes from 0 to 9:



The `\Note` command might also print several notes tied together with a beam. Two notes will be tied together with an oblique beam, but three or more will have a straight beam.

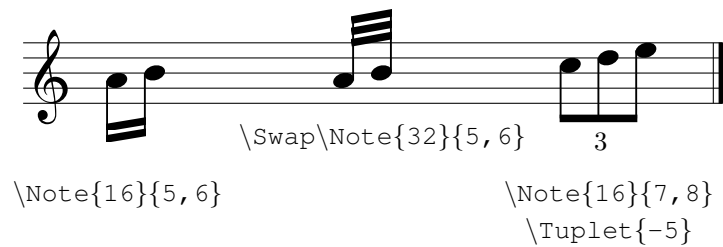
The notes go upside down when their height (or maximum height) is bigger than 6. However, for manual control, the `\Swap` command might be used *before* the note.

monotone
triplets slid-
ing beam

fix clip

To write the number of a tuplet, the `\Tuplet` command might be used *after* the note. It has two arguments, an optional one and a compulsory one. The optional one is the written number, which is automatically calculated by default. The compulsory one is the height.

bar on tu-
plet, fix
height



Notes with different durations that are tied together have a different syntax. Instead of using 8, 16 and 32 for the duration, each note has to be referenced with a -, a = or a ≡, respectively, and additional | might be needed for the edges.



`\Note{|≡,-,|=,=,|=,-,|=,≡|}{0,1,2,3,4,3,2,1}`

explain better

bigger space

To be able to write the ≡ as a symbol on the .tex file, I usually copy and paste it wherever I need it. Nonetheless, an alternative keyboard-friendly input that is equally valid is using /// instead of ≡.

3.5.2 \Rest

The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



If a multimeasure rest is needed —where the rest lasts an arbitrary amount of bars— an optional argument accepts a *.

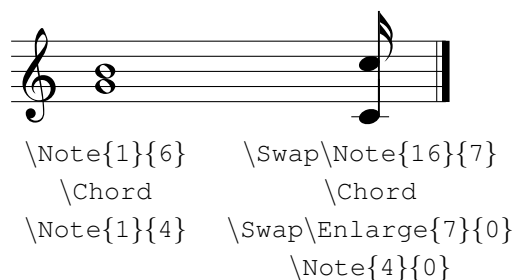


3.5.3 `\Chord`

Chords may be obtained by means of the `\Chord` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\Note` command— with 1 as default.

The best way to use it is to call the `\Note` command with the highest notes; then call the `\Chord` command; then call the `\Note` command with the duration 4 of a quarter note, to print them on top.

If the stem needs to be longer, the user may need the `\Enlarge` command, which accepts as arguments the 2 heights of both notes.

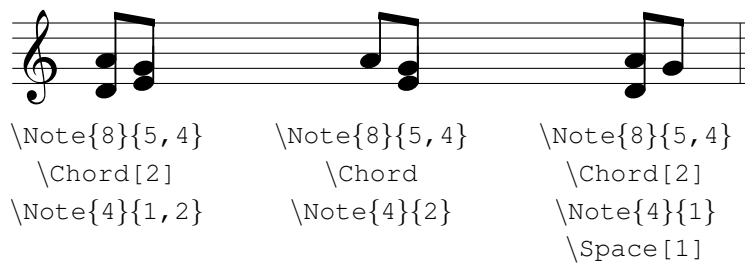


```

\Note{1}{6}      \Swap\Note{16}{7}
  \Chord          \Chord
\Note{1}{4}      \Swap\Enlarge{7}{0}
                  \Note{4}{0}

```

For extreme uses of the command, extra `\Space` commands might be needed.



```

\Note{8}{5,4}    \Note{8}{5,4}    \Note{8}{5,4}
  \Chord[2]       \Chord           \Chord[2]
\Note{4}{1,2}    \Note{4}{2}       \Note{4}{1}
                                   \Space[1]

```

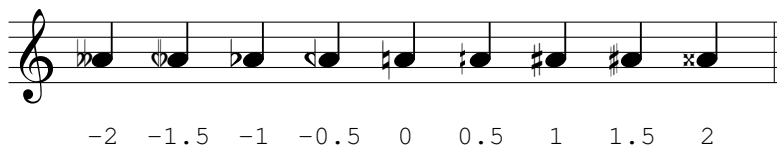
maybe fix
this

3.6 Keys and accidentals

3.6.1 `\Acc`

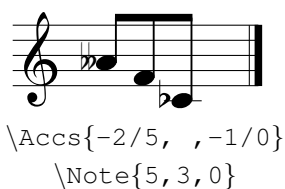
The `\Acc` command prints an accidental before a note. It accepts 2 arguments.

The first argument is the type of accidental.



The second argument is the height of the symbol.

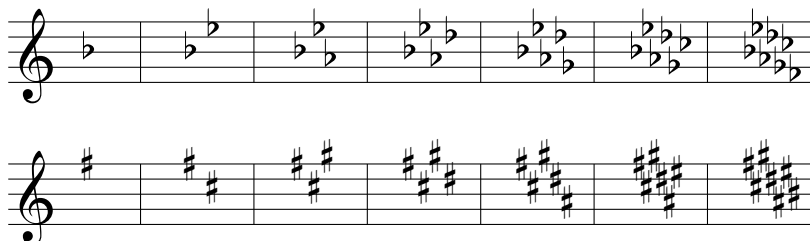
When there are several notes, the `\Accs` command might be useful. It accepts as an argument a list of pairs: type/height. Where there is no accidental, the spot shall be left empty.



3.6.2 `\Key`

The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps $\sharp$, and a negative whole number from -1 to -7 for flats $\flat$.



3.6.3 `\NoKey`

The `\NoKey` command is the same as the `\Key` command, but it prints naturals $\natural$ instead of sharps or flats.





Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```
\begin{mousik}
  \Clef[4]
  \Key{1}
  \TimeSig{C|}{}

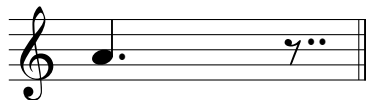
  \Acc{0}{8}
  \Note{1}{8}
  \Bar[|.]
\end{mousik}
```

3.7 Other symbols: dots, ties, slurs and text

3.7.1 `\Dot`

The `\Dot` command prints any amount of dots in front of a note or rest. It accepts an optional argument and a compulsory argument.

The default number of dots is 1, but it may be any whole number bigger than one. The second, compulsory argument is the height of the dot: `\Dot[2]{7}`.

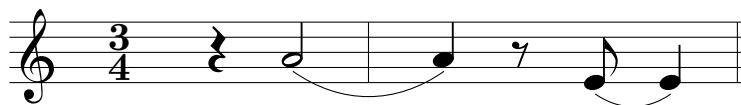


```
\Dot{5}      \Dot[2]{7}
\Note{4}{5}  \Rest{8}
```

3.7.2 `\Tied`

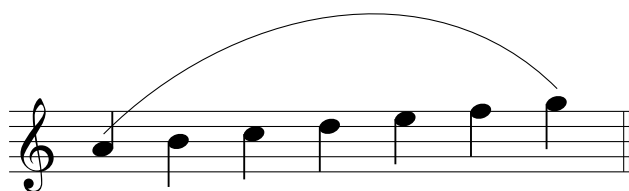
The `tied` environment draws a tie between the two notes placed inside. It has two arguments: the heights of both beginning and end notes (usually the same).

fix, add
swap



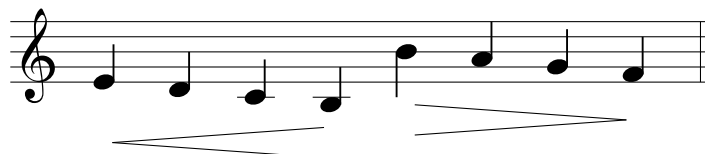
3.7.3 The **slurred** environment

The `slurred` environment draws a slur between the notes placed inside. It has two arguments: the heights of both beginning and end notes.



Fix angle

3.7.4 The **Crescendo** and **Decrescendo** environments



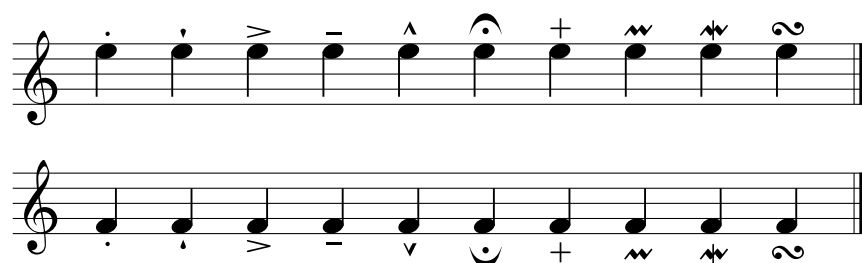
Explain
crescendo

Half line

3.7.5 Articulation

- Staccato: { . }
- Staccatissimo: { ' }
- Accent: { > }
- Tenuto: { - }
- Marcato: { ^ }
- Fermata: { (.) }
- Trill: { + }
- Mordent: { ~ }
- Lower mordent: { ~ | }

- Turn: {?}
- Snap: {o|}
- Open note: {o}
- Sull arco: {v}
- Giu arco: {n}
- Breath: {,}
- Caesura: {||}
- Simile: {\%
- Segno: {s\%
- Coda: {o+}
- Arpeggio: {s}
- Glissando: {/}

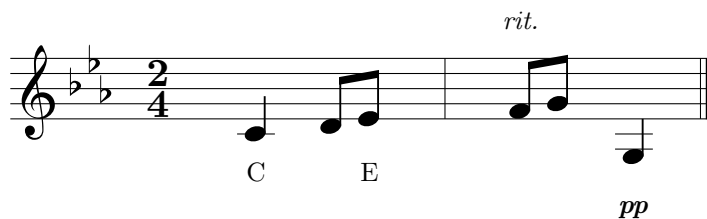


finish articulation

3.8 \Text and \Dynamics

The `\Text` command allows the printing of any text string. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.

To write anything in the usual font of dynamics, the `\Dynamics` command will work the same: `\Dynamics[-0.5]{pp}`.



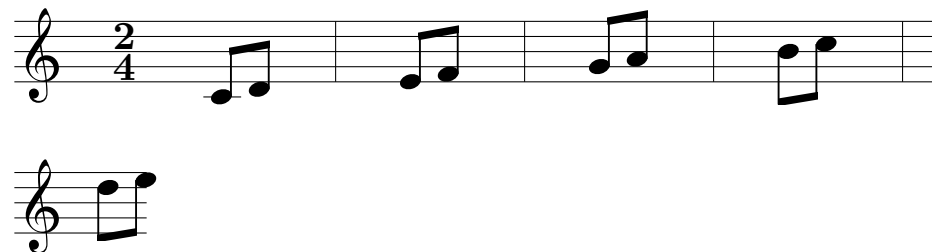
3.9 `\Tempo`

The `\Tempo` command prints a metronome tempo indicator. It has two arguments: the duration of the note and the numerical tempo which will be printed.



4 Tips and observations

4.1 Overflow of lines



4.2 How do spaces work?

Spaces serve as the manual progression of an invisible cursor during the printing. A negative

explain spaces

4.3 What if it becomes too cumbersome: your own commands

If a concrete rhythm is constantly used on a music piece.

4.4 On your own: working with the `tikz` package

explain tikz

5 A very long and complete example

$\text{♩} = 62$

pp

5

6

3

f